

PRE-CLASS READING 2 OF 5

Artificial Intelligence Index Report 2026

Stanford Institute for Human-Centered Artificial Intelligence (HAI). 9th edition, 425 pp.

ASSIGNED READING

Top Takeaways — read closely	pp. 9–11
Chapter 2 Highlights — Technical Performance — skim	p. 71
Chapter 4 Highlights — Economy — skim	pp. 173–174
≈ 7 pages · approx. 30 minutes	

Note on scope. The full report is 425 pages across nine chapters. You are reading the distilled findings only. Treat the rest as a reference work — when you need a defensible number for A1, A2 or your project, this is where you get it. Chapter map at the end of this sheet.

KEY TAKEAWAY

Capability is accelerating and adoption has gone mainstream faster than the PC or the internet — but the systems built around AI (evaluation, governance, education, disclosure) are not keeping pace. The co-chairs' framing: “*a field that is scaling faster than the systems around it can adapt.*” The operative consequence for you: published capability claims are getting harder to trust exactly as the stakes of relying on them rise.

WHY IT MATTERS FOR THIS COURSE

Mollick gives you the conceptual frame; the AI Index gives you the current numbers to hang on it. Takeaway #4 is the Jagged Frontier with 2026 data attached. The Economy chapter is the empirical base for Class 5. The Responsible AI findings set up Class 6. And when **A1** asks you to justify choosing a classifier over a foundation-model API, this report is where a defensible citation comes from.

MAIN POINTS

The frontier is jagged — now measured

- **Gold medal, can't read a clock.** Gemini Deep Think scored 35 points — gold — at the 2025 International Mathematical Olympiad, working end-to-end in natural language inside the 4.5-hour limit. On ClockBench, the top model reads analog clocks correctly **50.6%** of the time against a **90.1%** human baseline. The report calls this “**jagged intelligence.**” This is Mollick's Chapter 3 argument, independently measured a year later.
- **Agents leapt, then plateaued short.** On OSWorld (real computer tasks across operating systems), agent accuracy rose from ~12% to **66.3%** in one year — within 6 points of human performance, but still failing roughly **1 in 3** attempts. Directly relevant to what you can expect from your Class 4 agent build.
- **Robots remain the reality check.** **89.4%** success on RLBench in simulation; **12%** on real household tasks. The lab-to-world gap is the whole story of physical AI.
- **Benchmarks themselves are unreliable.** Invalid-question rates run from **2%** on MMLU Math to **42%** on GSM8K. Frontier models gained **30 percentage points in one year** on Humanity's Last Exam — a benchmark purpose-built to resist them. Benchmarks designed to last years now saturate in months.

The competitive landscape has converged

- **US-China gap effectively closed.** DeepSeek-R1 briefly matched the top US model in February 2025; as of March 2026 the top US model leads by **2.7%**. The US produced **59** notable models in 2025 to China's 35, and holds higher-impact patents — but China leads on publication volume, citations, patent grants and industrial robots.
- **Four labs inside 25 Elo points** (March 2026): Anthropic 1,503 · xAI 1,495 · Google 1,494 · OpenAI 1,481 · Alibaba 1,449 · DeepSeek 1,424. When models converge, competition shifts to **cost, reliability and domain fit** — the build-vs-buy analysis in Classes 2 and 5.
- **The open-weight gap reopened.** Closed leads open by **3.3%** (March 2026), up from just 0.5% in August 2024. Six of the Arena top ten are closed.

- **Industry dominates, disclosure is falling.** Industry produced **over 90%** of notable models in 2025. The Foundation Model Transparency Index average **fell from 58 to 40** in one year.
- **Small models are punching up.** OLMo 3.1 Think 32B matches Grok 4 on several benchmarks with **~90× fewer parameters**. A 111M-parameter protein model beat prior leading methods; a 200M-parameter genomics model outperformed one nearly **200× larger**. Bigger is not automatically better — direct input to your A1 model-selection reasoning.

Adoption is mainstream; value capture is uneven

- **Fastest-diffusing technology on record.** Generative AI reached **53%** population-level adoption within three years — faster than the PC or the internet. Organizational adoption hit **88%**.
- **But agents are barely deployed.** AI agent deployment is in **single digits** across nearly every business function — the opportunity space for your end-of-course project.
- **Investment more than doubled.** US private AI investment reached **\$285.9B** in 2025 versus **\$12.4B** in China — though the report cautions this understates China, whose government guidance funds deployed an estimated **\$184B** into AI firms (2000–2023).
- **Productivity gains are real but bounded.** **14–15%** in customer support, **26%** in software development, **50%** in marketing output — concentrated in structured, measurable work. The report flags emerging evidence of **long-term learning penalties** from heavy reliance.
- **The labour signal is narrow but sharp.** US software developers aged **22–25** saw employment fall nearly **20%** from 2024, while headcount for older developers grew.

Responsible AI is falling behind

- **Incidents rose to 362** in 2025, up from 233 in 2024.
- **Hallucination rates range 22–94%** across 26 top models. Models handle a false statement well when framed as another person's belief — but performance **collapses** when the same falsehood is framed as the user's belief. GPT-4o dropped from 98.2% to 64.4%; DeepSeek R1 from over 90% to 14.4%.
- **The dimensions trade off against each other.** Training techniques that improve one responsible-AI dimension consistently degraded others — a live constraint for your A2 guardrail checklist.
- **AI works best in English, and worse than benchmarks suggest.** On a Slovenian commonsense test, leading models lost close to **half** their accuracy in dialect versus standard language. Relevant if your project touches Chinese, Hindi, Bahasa, Tamil or any non-English deployment.

Your region, specifically

- **Singapore over-performs on adoption — 61%**, well above what GDP per capita predicts. **The United States ranks 24th, at 28.3%** — despite leading on investment and model development.
- **Southeast Asia is the world's most AI-optimistic region.** In Malaysia, Thailand, Indonesia and Singapore, **over 80%** say AI will profoundly change their lives within 3–5 years.
- **Workplace use is highest in emerging economies.** 58% of employees globally use AI at work — but in **India, China, Nigeria, UAE and Saudi Arabia** the share exceeds **80%**.
- **India registered the sharpest rise in AI nervousness** of any country surveyed: **+14 points** between 2024 and 2025, against only +2 on excitement.
- **Singapore reports the highest trust in its own government to regulate AI, at 81%** (global average 54%). The United States reports the lowest, at 31%.
- **Singapore and Switzerland lead the world in AI researchers per capita.** NTU appears in the report's institutional ranking for the 100 most-cited AI papers.
- **China installed 54%** of the world's industrial robots in 2024 — more than the rest of the world combined.

Experts and the public disagree, sharply

- On how people do their jobs: **73%** of experts expect a positive impact versus **23%** of the public — a **50-point gap**. Economy: 69% vs 21%. Medical care: 84% vs 44%.
- **64%** of Americans expect AI to lead to fewer jobs over 20 years; only **5%** expect more.

READING FOCUS QUESTIONS — bring your answers to class

There are no wrong answers here — these are meant to surface how you think, not to test recall.

1. Takeaway #4 says a model can win an Olympiad gold medal in math but read an analog clock correctly only about half the time. Think of something about *yourself* that works the same way — a skill you're weirdly good at that has nothing to do with how hard it looks from outside, or something that looks easy but you're surprisingly bad at. What does that tell you about how "hard" something looks versus how hard it actually is — for you, or for a model?
2. 88% of organizations use AI somewhere, but AI agents — systems that act on their own — are barely deployed anywhere. Think about apps or services you already use that could, in theory, act on your behalf right now (booking something, replying to messages, managing your calendar) but mostly don't yet. What do you think is actually holding that back — the technology not being reliable enough, people not trusting it, or something else?
3. Singapore is at 61% adoption and the US at 28.3%, despite the US leading investment and model development. What explains the inversion? (If you happen to know a similar figure for your own home country, factor that in too.)

CHAPTER MAP (for later reference — you are not reading these now)

Ch.	Title	p.	Go here for
1	Research and Development	12	Model counts, compute, chips, talent flows, environmental footprint
2	Technical Performance	68	Benchmarks, agents, robotics, open vs. closed
3	Responsible AI	126	Incidents, safety evals, transparency, bias — A2 guardrails
4	Economy	171	Investment, adoption, productivity, labour — Class 5
5	Science	231	AI in research workflows (new this edition)
6	Medicine	255	Clinical AI, ambient scribes (new this edition)
7	Education	288	Student and teacher AI use
8	Policy and Governance	323	EU AI Act, national strategies, AI sovereignty — Class 6
9	Public Opinion	360	Global sentiment, trust, expert-public divergence

SOURCE LINKS

Report landing page (chapters + downloads) · hai.stanford.edu/ai-index/2026-ai-index-report

AI Index home · aiindex.stanford.edu

Full report PDF · also supplied in your **pre-class-readings** folder as `ai_index_report_2026.pdf`

CAVEATS

1. **This summary is not a substitute for the reading.** It is a map, not the territory. Read the Top Takeaways and the two chapter highlight pages — and return to the full report as a reference throughout the course.
2. **This summary was generated and fact-checked using Anthropic models.** Every figure above was extracted directly from the report PDF. Errors are still possible — the report is the authority. Flag anything that looks wrong.